# Halting and settling: why the shells do not stop by themselves, and what the settling into a core requires

B. R. Parodi

companion note to *From a primordial fluctuation to an idealised rotation curve* (pf2irc), assumptions (A3) and (A4), Sect. 8.1 and Appendix H

29 October 2026

**Abstract**

The paper builds a dwarf galaxy from a seed whose shells share one energy, $E = -v^2/2$, with two assumptions: each shell halts at the accretion front (A3), and the halted matter settles into the cored P18 profile (A4). This note asks what these assumptions demand of real dynamics. *Part I, halting.* The halting radius is the radius where gravity and centrifugal force balance, and halting there is a turn of the velocity at fixed speed: it needs no energy, but ten times the angular momentum that the seed's spin supplies, and even the right angular momentum would leave the shells oscillating between 0.73 and $2\,R_b$. Gas can halt by shocks and cooling; the dark matter only by violent relaxation and non-spherical torques. Star formation and supernovae cannot stop the dark matter: they act on the gas, far inside the front, and supply energy rather than angular momentum. Test particles in potentials with broken symmetry show that a flattening about one axis turns the infall but cannot keep the turn, and that a triaxial exponential sphere, the component that carries the mass excess, cannot halt the shells but partly flattens the central cusp of radial orbits. *Part II, settling.* P18 crosses the singular isothermal sphere (SIS) exactly at $r = r_c$; it is the SIS with $0.52\,v^2 r_c/G$ moved from inside $r_c$ to between $r_c$ and about $5r_c$, only 0.3–1.5% of a dwarf's mass. Distribution functions show that P18 can be built from orbits that are nearly radial outside about $2r_c$, while inside $r_c$ they must be close to isotropic (the most radial Osipkov–Merritt model has $r_a = 0.87\,r_c$), and that the most bound matter of the SIS must be lifted by an energy of order $v^2$. Settling therefore means isotropising the orbits in the core and raising its energy, continuously while the core grows, through a potential that changes in time; it does not mean circularising the shells at the front. All derivations are given step by step, and all numbers come from the scripts of this package.

## Contents

**I Halting** **2**

**1 The seed and the halting rule in brief** **2**

**2 Why the shells do not halt by themselves** **3**

The end state is in force balance . . . 3
The halting radius is the radius of balance . . . 3
The halt is a turn, not a stop . . . 3
The seed gives ten times too little . . . 4
Even the right angular momentum is not enough . . . 4
What could make the shells settle . . . 4
Can star formation and supernovae halt the shells? . . . 6
Harmonic motion: what a settled shell does . . . 7
What this means for the model . . . 8

**3 Breaking the symmetry: two numerical experiments** **8**
1. A flattening about one axis . . . 8
2. A triaxial exponential sphere . . . 9
What the two experiments show . . . 10

**II Settling into a core** **10**

**4 The question** **10**

**5 Where the mass is: P18 as a rearranged SIS** **11**
5.1 The profile . . . 11
5.2 A useful rewriting of $u$ . . . 11
5.3 How much mass is moved . . . 12
5.4 A remark on the very centre . . . 12

**6 Circular orbits: the angular momentum of each shell** **12**

**7 Orbit populations: distribution functions** **13**
7.1 What a distribution function is . . . 13
7.2 Eddington's inversion, step by step . . . 14
7.3 The isotropic P18 . . . 14
7.4 How radial may the orbits be? . . . 15

**8 How tightly the mass is bound** **15**

**9 What this means for assumption (A4)** **16**
9.1 What (A4) requires . . . 17
9.2 What (A4) does not require . . . 18
9.3 The candidate mechanisms, measured against these requirements . . . 18
9.4 What a simulation should look for . . . 19

**10 Limits of this note** **19**

# Part I
# Halting

## 1 The seed and the halting rule in brief

With the common energy $E = -v^2/2$ and a common bang time $t = 0$, the shell enclosing the mass $M$ follows the cycloid

$$R = \frac{GM}{v^2}\,(1 - \cos\eta), \qquad t = \frac{GM}{v^3}\,(\eta - \sin\eta). \tag{1}$$

It turns around at $R_{\rm ta} = 2GM/v^2$ and, by the halting rule (A3), halts on its way back where the galaxy already built encloses its mass: at $R \simeq GM/v^2$, i.e. at the phase $\eta_b = 3\pi/2$. The halting points lie on the accretion front $R_b = vt/(1 + 3\pi/2) \simeq 0.175\,vt$, and behind it $M \propto R_b$ gives a flat rotation curve. With a cosmological constant the cycloids are slightly modified and the front lags by a few per cent (paper, Sect. 3.4 and Appendix A). Pictures of the nested top hats at one and at two moments, and the density profile of the seed, are given in the companion note *Two pictures of one top hat*. Part I asks why a shell does not halt by itself, and what could make it settle; Part II asks what the settled matter must look like in phase space.

## 2 Why the shells do not halt by themselves

In the model a shell stops falling at the front, $R_b \simeq GM/v^2$, and stays there (assumption A3). Real matter is not stopped by a wall. Could a shell halt by itself, through the balance of gravity and centrifugal force? This section shows that the halting radius is indeed the radius of such a balance, but that a shell with the seed's angular momentum cannot reach it on its own.

**The end state is in force balance.** In the settled galaxy every particle moves on a circular orbit, its orbital plane oriented at random, so that each spherical shell stays spherical and the whole has no net rotation: an *Einstein cluster*. For every particle gravity and centrifugal force balance,

$$\frac{v_c^2}{R} = \frac{GM(R)}{R^2}, \qquad v_c^2 = \frac{GM(R)}{R} = v^2 \tag{2}$$

in the flat part of the curve. (The paper uses the virial theorem, which gives the same kinetic energy for any steady state, but the Einstein cluster is the simplest picture of it.) The question is therefore not *whether* the end state is in force balance (it is, by construction), but how a falling shell gets into it.

**The halting radius is the radius of balance.** Take the galaxy already built inside the front as a singular isothermal sphere with flat speed $v$, so that $M(<R) = v^2R/G$ inside the front. The shell of mass $M$ then feels the potential

$$\Phi(R) = \begin{cases} v^2 \ln(R/R_b) - v^2, & R \le R_b, \\ -GM/R, & R > R_b, \end{cases} \qquad R_b = \frac{GM}{v^2}, \tag{3}$$

continuous at the front, where both equal $-v^2$. A circular orbit at a radius $R \le R_b$ inside the galaxy has the speed $v$ (the curve is flat) and therefore the energy

$$E_{\rm circ}(R) = \tfrac{1}{2}v^2 + \Phi(R) = v^2\Big[\ln\frac{R}{R_b} - \frac{1}{2}\Big]. \tag{4}$$

At the front this is

$$E_{\rm circ}(R_b) = \tfrac{1}{2}v^2 - v^2 = -\tfrac{1}{2}v^2, \tag{5}$$

*exactly the energy of the seed's shells.* Inside the front circular orbits are more bound ($E_{\rm circ} = -1.19\,v^2$ at $R_b/2$), outside less bound (Kepler orbits with $E_{\rm circ} = -GM/2R$, $-0.25\,v^2$ at $2R_b$). Since $E_{\rm circ}$ rises monotonically with $R$, the front is the *only* radius at which a shell with the seed's energy can move on a circular orbit, i.e. be in centrifugal balance. The halting rule (A3) picks this radius for every shell, because every shell has the same energy. On that orbit the shell needs the specific angular momentum

$$j_{\rm circ} = vR_b = \frac{GM}{v}. \tag{6}$$

For the example galaxy of the paper ($v = 60\,\mathrm{km\,s^{-1}}$, growth ending at $z = 0.5$, $R_b = 84.2\,\mathrm{kpc}$) the outermost shell would need $j_{\rm circ} = 5050\,\mathrm{kpc\,km\,s^{-1}}$, and its circular orbit at $R_b$ would take $2\pi R_b/v = 8.6\,\mathrm{Gyr}$, about the age of the galaxy at the end of growth.

**The halt is a turn, not a stop.** A shell falling radially arrives at $R_b$ with the speed $|\dot R| = v$ (paper, Appendix C): its kinetic energy there, $\frac{1}{2}v^2$, is all radial. On the circular orbit at $R_b$ it would have the same kinetic energy, $\frac{1}{2}v^2$, all tangential. Halting a shell at the front therefore means *turning its velocity by 90 degrees at fixed speed.* Its energy does not change, which is why the model can conserve energy shell by shell; what must change is its angular momentum, from what the seed gave it to $GM/v$.

**The seed gives ten times too little.** In the paper each shell receives at turnaround, where $R_{\rm ta} = 2R_b$, the angular momentum $j = \sqrt{2}\lambda v R_{\rm ta} = 2\sqrt{2}\lambda\, GM/v$. This adds the rotational energy $j^2/2R_{\rm ta}^2 = j^2/8$ (in units $G = M = v = 1$), so the shell moves with $E = -\frac{1}{2} + j^2/8$ and the angular momentum $j$ in the effective potential

$$\Phi_{\rm eff}(R) = \Phi(R) + \frac{j^2}{2R^2}, \qquad \tfrac{1}{2}\dot{R}^2 = E - \Phi_{\rm eff}(R). \tag{7}$$

Its turning points are the roots of $E = \Phi_{\rm eff}(R)$: the apocentre is the turnaround radius $2R_b$, and the pericentre $R_p$ lies deep inside the galaxy. Table 1 gives them for the spread of halo spins ($\lambda = 0.035$ is the median; 0.021 and 0.058 bracket 68% of a lognormal with $\sigma_{\ln\lambda} = 0.5$; 0.07 is twice the median) and for the spin that would give $j_{\rm circ}$, $\lambda = 1/(2\sqrt{2}) = 0.354$.

**Table 1:** Orbits of a shell with the seed's energy in the frozen galaxy, Eq. (3) (units $G = M = v = 1$, $R_b = 1$; `halt_numbers.py`). $T_r$: radial period, from apocentre to apocentre.

| $\lambda$ | $j$ $[GM/v]$ | $E$ $[v^2]$ | pericentre $R_p$ | $T_r$ $[GM/v^3]$ |
|---|---|---|---|---|
| 0.021 | 0.059 | −0.4996 | $0.020\,R_b = R_b/50$ | 6.47 |
| 0.035 | 0.099 | −0.4988 | $0.036\,R_b = R_b/28$ | 6.48 |
| 0.058 | 0.164 | −0.4966 | $0.064\,R_b = R_b/15.5$ | 6.53 |
| 0.070 | 0.198 | −0.4951 | $0.081\,R_b = R_b/12.4$ | 6.55 |
| 0.354 | 1.000 | −0.3750 | $0.73\,R_b$ | 9.37 |

For every realistic spin the angular momentum is 5–20% of $j_{\rm circ}$, and the orbit is extremely eccentric: from $2R_b$ to between $R_b/50$ and $R_b/12$. The centrifugal barrier, $j^2/2R^2$, rises only where $R \lesssim j\,GM/v^2$, i.e. at a few per cent of $R_b$; there the shell is turned round for an instant, and it swings back out (Fig. 1). For comparison: a circular orbit with the seed's $j = 0.099$ would lie at $R = j\,R_b = 0.099\,R_b$ and have $E_{\rm circ} = -2.81\,v^2$ by Eq. (4); the shell carries $2.31\,v^2$ more than that, all in radial motion. A collisionless shell keeps its energy and angular momentum, so the oscillation goes on: the radial period, $6.5\,GM/v^3$, is 8.9 Gyr for the outermost shell of the example galaxy ($GM/v^3 = 1.37$ Gyr), and correspondingly less for inner shells. (In reality the galaxy keeps growing, which pulls the apocentres in to about $1.2$–$1.4\,R_b$, the splashback radius of the paper, Sect. 3.3; it does not stop the oscillation.)

**Even the right angular momentum is not enough.** Suppose the shell received at turnaround the full $j_{\rm circ} = GM/v$ (the run $f = 1$ of the paper, Appendix H). It then has the angular momentum of the circular orbit at $R_b$, but not its energy: receiving $j$ at turnaround, where the shell is momentarily at rest, adds the rotational energy $j^2/2R_{\rm ta}^2 = v^2/8$, so that $E = -\frac{3}{8}v^2$ instead of $-\frac{1}{2}v^2$. The shell then oscillates between $0.73\,R_b$ and $2\,R_b$ (Fig. 1b, red). A circular orbit requires the right angular momentum *and* the right energy; equivalently, the rotation must come at the expense of the radial motion, not on top of it. (This is the alternative booking of the spin energy discussed in the paper, Sect. 4.2 and Table 1: "spin energy inside $-v^2/2$".)

**What could make the shells settle.** Settling requires that individual particles change their energy and their angular momentum, while the totals are conserved. For the outermost shell of the example galaxy, the angular momentum per unit mass must grow from about 500 to 5050 kpc km s$^{-1}$, and the radial kinetic energy at the front, $\frac{1}{2}v^2 = 1800\,(\mathrm{km\,s^{-1}})^2$, must be turned into tangential motion. Three processes can do this.

- *Dissipation, for the gas.* Gas falling at $v$ is stopped by an accretion shock near the front (cf. Bertschinger 1985). A strong shock heats it to

$$T = \frac{3\mu m_p}{16k}\,v^2 \simeq 4.8\times10^4\,\mathrm{K} \quad (v = 60\,\mathrm{km\,s^{-1}};\ 1.3\times10^5\,\mathrm{K\ for\ 100\,km\,s^{-1}}),$$

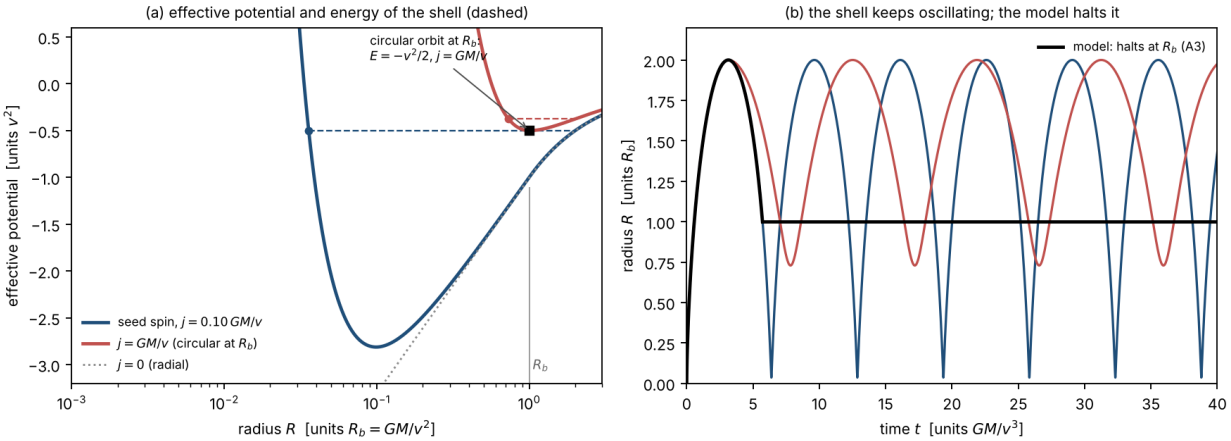



**Figure 1:** One shell (mass $M$, energy $-v^2/2$) falling into the galaxy already built, taken as a singular isothermal sphere frozen at the moment of arrival; units $G = M = v = 1$, so that $R_b = 1$. **(a)** Effective potential $\Phi + j^2/2R^2$ for the seed's spin, $j = 0.099\,GM/v$ (blue), for $j = GM/v$ (red) and for $j = 0$ (dotted), with the shell's energy (dashed) between pericentre (dot) and apocentre. The square marks the circular orbit at $R_b$, which has exactly the seed's energy. **(b)** Radius against time: radial cycloid up to turnaround, then the orbit with the seed's spin (blue, pericentre $R_b/28$) or with $j = GM/v$ added at turnaround (red, between 0.73 and $2\,R_b$); black: the model, which halts the shell at $R_b$. Script: `halt.py`, numbers in `halt_out.txt`.

with $\mu = 0.59$. This is close to the peak of the cooling curve of ionised gas near $10^5$ K, so the gas radiates the energy quickly and settles, with the angular momentum it has, into a disc. For the baryons, halting is natural, and the energy they radiate is the cooling term of the paper (Sect. 5.4).

- *Violent relaxation, for the dark matter.* In a collapsing system the potential changes on the dynamical time, and a particle's energy is not conserved in a time-dependent potential: the rate of change is $\dot{E} = \partial\Phi/\partial t$ along its orbit. The energies are thus reshuffled among the particles, with changes of order $v^2$ during the collapse (Lynden-Bell 1967). In exact spherical symmetry, however, the force is radial and each particle's angular momentum is conserved, so violent relaxation alone cannot circularise the orbits.

- *Torques from departures from spherical symmetry.* A potential with a non-spherical part of relative amplitude $\varepsilon$ exerts a tangential force of order $\varepsilon\,GM/R^2$, i.e. a torque per unit mass $\varepsilon\,GM/R$. Over a dynamical time $R/v$ this changes the angular momentum by

$$\Delta j \sim \varepsilon\,\frac{GM}{R}\,\frac{R}{v} = \varepsilon\,\frac{GM}{v}.$$

  Raising $j$ from 0.1 to $1\,GM/v$ in a few dynamical times thus requires $\varepsilon$ of order 0.3–1: strong, not slight, departures from spherical symmetry during the collapse. A cold, nearly radial collapse does produce them: it is unstable to bar-like and triaxial distortions (the radial-orbit instability; MacMillan et al. 2006), and simulated haloes are triaxial with axis ratios of about 0.5–0.8. Since the orbits end up oriented at random, each particle can gain an angular momentum of order $GM/v$ while the total stays small, as in an Einstein cluster; this is why the halo can be nearly isotropic although its net spin is only $\lambda \simeq 0.035$.

None of these exists in a spherical shell code with conserved angular momenta, which is why the paper's collisionless run (Appendix H) neither halts its shells nor forms a core. The $f = 1$ run comes closest to the effect of the torques: it gives the shells the angular momentum of circular orbits, though on top of their energy, and then yields a nearly flat, isothermal curve, but still no core.

**Can star formation and supernovae halt the shells?** Collapsing gas becomes dense, forms stars, and the massive ones explode and release energy. One may ask whether this feedback could stop the infall at the front. The answer is no for the dark matter, which is what must halt, for four reasons; feedback can, however, stop the gas and may help to make the core. The numbers below are for the example galaxy and are computed in `sf_numbers.py`.

1. *Feedback acts on the gas, not on the dark matter.* The energy of a supernova, $10^{51}$ erg, goes into heat and momentum of the surrounding gas. Dark matter is collisionless and feels neither pressure nor shocks; it responds only to gravity. A dark-matter shell can therefore be affected by feedback only indirectly, through a change of the gravitational potential, and only by the amount $\dot{E} = \partial\Phi/\partial t$ along its orbit.

2. *Feedback happens in the wrong place.* Massive stars form in dense, cold gas, at hydrogen densities above about 0.1–10 $\mathrm{cm}^{-3}$, i.e. in the inner disc, within a few kpc of the centre (the example galaxy has $r_c = 2.44\,\mathrm{kpc}$ and $R_d \simeq 0.67\,r_c \simeq 1.6\,\mathrm{kpc}$). The shells halt at the front, $R_b = 84\,\mathrm{kpc}$. There the mean density inside the front is 147 times the cosmic mean, $1.4 \times 10^{-27}\,\mathrm{g\,cm}^{-3}$ for growth ending at $z = 0.5$, and the local density at the front is a third of that, $4.6 \times 10^{-28}\,\mathrm{g\,cm}^{-3}$. Even with the full cosmic baryon fraction (0.16) the gas there has $n_\mathrm{H} \simeq 3 \times 10^{-5}\,\mathrm{cm}^{-3}$ ($8 \times 10^{-5}$ at $z = 1$, $3 \times 10^{-4}$ at $z = 2$): two and a half to five and a half orders of magnitude below the threshold for star formation ($0.1/3 \times 10^{-4} \simeq 300$ at the closest). No stars form where the shells have to halt.

3. *The indirect effect at the front is small.* Moving gas changes the potential at the front by at most the fraction of the mass it moves: if a fraction $f$ of the mass inside $R_b$ is carried outside, $\Phi(R_b) = -v^2$ changes by at most $f\,v^2$. Our dwarfs hold baryons amounting to only $f \simeq 0.017$ of the total mass (paper, Sect. 5.4), so a shell's energy changes by at most about $0.02\,v^2$ per such event, against the $\frac{1}{2}v^2$ of radial motion that has to be turned into tangential motion. Even if all the cosmic baryons, $f = 0.16$, were blown out at once, the change would be $0.16\,v^2$, and it would still be an energy change: a spherically symmetric outflow exerts no torque, so the shell's angular momentum, and hence its pericentre at $R_b/28$, would stay the same.

4. *Feedback supplies energy, but the halt needs angular momentum.* As shown above, halting a shell at $R_b$ requires no energy at all, only a turn of its velocity, i.e. an increase of its angular momentum from about 0.1 to $1\,GM/v$. Feedback adds energy: it makes matter less bound and pushes it outwards. It is not small: one supernova per $100\,M_\odot$ of stars formed releases $(709\,\mathrm{km\,s}^{-1})^2$ per unit stellar mass, and if a fraction $f_c$ of it couples to the matter, the energy gained in units of the binding energy $\frac{1}{2}v^2M$ is

$$\epsilon_\mathrm{fb} = 2f_c\Big(\frac{709\,\mathrm{km\,s}^{-1}}{v}\Big)^2\frac{\Delta M_\star}{M} \simeq 280\,f_c\,\frac{\Delta M_\star}{M} \quad (v = 60\,\mathrm{km\,s}^{-1}),$$

   a median of $0.65\,f_c$ for the 79 dwarfs (paper, Table 2). But it is the wrong kind of change: it alters how bound the matter is, not how its motion is divided between radial and tangential.

What feedback *can* do is different, and both effects are part of the paper's picture.

- *It can halt the gas.* Winds driven by supernovae, together with the heating by reionisation, can stop or reverse the accretion of gas onto small haloes. This is the most likely reason why our dwarfs hold only about a tenth of the cosmic baryon fraction, the "missing baryons" of the paper (Sect. 8.3). For the baryons, feedback can act as a halt; for the dark matter it cannot.

- *It can make the core.* In the centre, where the baryons are a substantial part of the mass (a median 24% of $V^2$ at $r_c$ in our dwarfs, 12–45% for the central 68%), repeated bursts that blow gas out and let it fall back make the central potential fluctuate on about a dynamical time. Each fluctuation changes the energies of the dark-matter particles there; because the particles lose energy while the potential is deep and gain it when it is shallow, the net effect

over many cycles is to heat the central dark matter and push it outwards, turning a cusp into a core (Navarro, Eke & Frenk 1996; Read & Gilmore 2005; Governato et al. 2010; Pontzen & Governato 2012). In the language of the paper, this is a candidate for the settling into a core (A4), not for the halt (A3).

- *It has a price in the model.* Feedback per unit binding energy falls steeply with the flat speed, as $v^{-2.5}$, whereas the energy radiated by the cooling gas falls only as $v^{-0.6}$ (paper, Sect. 5.4). Cores made by feedback after growth would therefore not follow the observed $r_c \propto v$ unless feedback and cooling nearly cancel in every dwarf, the fine-tuning of the paper. A feedback core is compatible with the model only if it forms during assembly or is regulated by cooling.

In short, the halting of the dark matter at the front must come from gravity alone: violent relaxation together with the torques of non-spherical distortions. Star formation and supernovae, which act later and much further in, can remove most of the gas and may shape the core, but they cannot stop the infall of the dark matter.

**Harmonic motion: what a settled shell does.** Once a shell is close to its circular orbit, its remaining motion is a set of harmonic oscillations. Three harmonic equations appear.

1. *Epicycles about the circular orbit.* Write $R = R_c + X$ with $X \ll R_c$ and expand $\Phi_{\rm eff}$, Eq. (7), about its minimum $R_c$, where $j^2 = R_c^3\,\mathrm{d}\Phi/\mathrm{d}R$. To first order

$$\ddot{X} + \kappa^2 X = 0, \qquad \kappa^2 = \frac{\mathrm{d}^2\Phi_{\rm eff}}{\mathrm{d}R^2}\Big|_{R_c} = R\frac{\mathrm{d}\Omega^2}{\mathrm{d}R} + 4\Omega^2, \tag{8}$$

   with $\Omega = v_c/R$ the angular frequency of the circular orbit. The solution is

$$X = A\cos(\kappa t + \alpha), \qquad \varphi = \Omega t - \frac{2\Omega}{\kappa}\,\frac{A}{R_c}\sin(\kappa t + \alpha), \tag{9}$$

   the second from $\dot{\varphi} = j/R^2 \simeq \Omega(1 - 2X/R_c)$. In the flat part of the curve $\Omega = v/R$ and $\kappa = \sqrt{2}\,\Omega$: the shell oscillates radially $\sqrt{2}$ times per revolution, so its orbit is a rosette that never closes. At the front of the example galaxy $2\pi/\Omega = 8.6\,\mathrm{Gyr}$ and $2\pi/\kappa = 6.1\,\mathrm{Gyr}$. The approximation holds only for $A \ll R_c$; the shells of the seed, with excursions from $2R_b$ to $R_b/28$, are far outside it, which is the quantitative sense in which they are far from balance.

2. *The harmonic core.* Near the centre of the P18 profile the density is constant, $\rho(0) = v^2/(\pi G r_c^2)$, so the potential is quadratic, $\Phi = \frac{2}{3}\pi G\rho(0)R^2 + \text{const}$, and each Cartesian coordinate obeys

$$\ddot{x} = -\omega^2 x, \qquad \omega^2 = \frac{4\pi G\rho(0)}{3} = \frac{4}{3}\,\frac{v^2}{r_c^2}, \tag{10}$$

   with the solution $x = a\cos(\omega t + \alpha)$, $y = b\cos(\omega t + \beta)$, $z = c\cos(\omega t + \gamma)$: centred ellipses, all with the same period $2\pi/\omega$ whatever their size. The circular speed rises linearly, $v_c = \omega R$, which is the inner rise $(2/\sqrt{3})(v/r_c)R$ of the idealised rotation curve; here $\kappa = 2\Omega$ and the orbits close. For the example galaxy $\omega = 28.4\,\mathrm{km\,s^{-1}\,kpc^{-1}}$ and $2\pi/\omega = 216\,\mathrm{Myr}$. A particle on a radial orbit through the core simply oscillates through the centre with this period.

3. *The cycloid itself.* Even the infall of a shell before it halts is a harmonic oscillator in disguise. With the time variable $\eta$ of Eq. (1), $\mathrm{d}t = (R/v)\,\mathrm{d}\eta$, the radial Kepler equation becomes

$$\frac{\mathrm{d}^2R}{\mathrm{d}\eta^2} + R = \frac{GM}{v^2}, \tag{11}$$

   whose solution starting at $R = 0$ is $R = (GM/v^2)(1 - \cos\eta)$: a harmonic oscillation about $GM/v^2 = R_b$, the halting radius, with amplitude $R_b$. In this variable the front is simply the point where the oscillation passes its centre for the second time, $\eta = 3\pi/2$. A cosmological constant adds a term $\propto R^3$ in this equation and breaks the harmonic form, which is why the lag of the front needs the perturbation theory of the paper's Appendix A.

**What this means for the model.** Assumptions (A3) and (A4) do not replace the force balance; they *assume* it. They place every shell at the radius where, with the seed's energy, gravity and centrifugal force balance, and they assume that the non-spherical relaxation of a real collapse has turned its radial motion into tangential motion there. Because energy is conserved in this turn, the energy bookkeeping of the paper does not depend on how the relaxation proceeds, only on its end state, just as the virial radius of the classical top-hat model does not depend on the details of virialisation. What the model cannot explain without such a mechanism is the core: as the core grows in proportion to $t$, the matter inside it must keep being rearranged (paper, Appendix I). Part II shows what this requires: nearly isotropic orbits in the core and an energy gain of order $v^2$ for its most bound matter. Whether non-spherical dynamics can provide this is the open problem named in the paper's Outlook (Sect. 8.3). The next section tests two simple ways of breaking the spherical symmetry.

## 3 Breaking the symmetry: two numerical experiments

Both experiments follow test particles in a fixed, prescribed potential, to which a non-spherical part is added that grows with time. They do not compute how the distortion arises; they test what it can do to the orbits once it exists. Units are $G = v = 1$. The scripts and their outputs are part of this package; Part II (Sect. 9) measures the results against what the settling requires.

**1. A flattening about one axis (`axi_test.py`, `axi_energy.py`).** Take the flat-curve potential of the settled galaxy and flatten it along $z$,

$$\Phi = \frac{v^2}{2} \ln\Big(r_s^2 + x^2 + y^2 + \frac{z^2}{q^2}\Big), \tag{12}$$

with a small softening $r_s = 0.01\, R_b$ and an axis ratio $q \leq 1$ of the potential, either fixed or decreasing from 1 to $q_{\min}$ over a time $T$. (Below $q \simeq 1/\sqrt{2}$ this potential would require a negative density at the poles, so $q = 0.7$–$0.8$ is the strongest flattening possible.) A particle starts at rest at the turnaround radius $2R_b$, at latitude 20°, 45° or 70°, and falls in. Its angular momentum $j$ is measured in units of $GM/v = vR_b$, the value of a circular orbit at $R_b$.

- *The flattening turns the infall.* It pulls the particle towards the equatorial plane, so radial motion is turned into motion along the meridian. At its first pericentre a particle at 45° has $j = 0.25$ for $q = 0.9$, 0.48 for $q = 0.8$ and 0.69 for $q = 0.7$; over several orbits $j$ reaches about 1. Particles at $+\lambda$ and $-\lambda$ are turned in opposite senses, so the result is a dispersion of tangential velocities, not a net rotation.

- *But the turn is not kept.* In any potential symmetric about the $z$-axis, static or not, $\partial\Phi/\partial\varphi = 0$, so the component $L_z$ of each particle's angular momentum is conserved. For the infalling particle $L_z = 0$, and it stays in its meridional plane. The angular momentum gained there is the component perpendicular to that plane, which is not conserved: the orbit precesses towards the equator and back, and $j$ swings up and down. The late pericentres are 0.02–$0.18\, R_b$: the orbits keep returning to the centre. Only $L_z$ could hold a particle away from the centre for good, and it comes only from the seed's spin, which is ten times too small.

- *A growing flattening adds energy.* As $q$ falls, the potential above and below the plane rises under the particle, which gains 0.01–$0.31\, v^2$ (`axi_energy_out.txt`); its apocentres stay at 2.0–$2.3\, R_b$ instead of moving in to $R_b$.

- *A slow growth comes too late.* If $q$ falls from 1 to 0.7 over 10 or 30 units of $R_b/v$, the first infall gains only $j = 0.06$–0.09 or 0.02–0.03; the turn comes orbits later.

A flattening about one axis thus cannot halt the shells at $R_b$. What is needed is a torque about the axis itself, which requires a distortion without any axis of symmetry: a triaxial or bar-like shape.

**2. A triaxial exponential sphere (`triax_core.py`).** The P18 density splits into an isothermal-like part, with $v_c = v(1-e^{-x})$, and an exponential sphere, $\rho_e = v^2 e^{-2R/r_c}/(4\pi G r_c^2)$, of mass $v^2 r_c/4G$, which carries the whole mass excess $c = \frac{1}{4}$ and a quarter of the central density (paper, Sect. 2.2). Suppose this central component becomes triaxial. Its potential is written as a function of the ellipsoidal radius,

$$\Phi = \Phi_1(r) + \Phi_e(m), \qquad m^2 = X^2 + \frac{Y^2}{p^2} + \frac{Z^2}{q^2}, \tag{13}$$

where $\Phi_1$ is the potential of the isothermal-like part, $\Phi_e$ that of the spherical exponential sphere, and $(X, Y, Z)$ are coordinates fixed in the distorted body, which either stays fixed or tumbles about its short axis at a pattern speed $\Omega_p$. The axis ratios $p, q$ of this potential grow linearly from 1 to their final values within $20\, r_c/v$ (0.8 Gyr for the example galaxy). For $p, q = 0.8, 0.6$ the total density has axis ratios of about 0.7–0.85 near $r_c$ and is positive everywhere (`triax_density_check.py`): a moderate distortion. Units are $r_c = 1$ and $R_b = 34\, r_c$, the example galaxy ($r_c = 2.44$ kpc, $R_b = 84$ kpc, $r_c/v = 40$ Myr).

The particles represent the matter accreted early, which in the paper's shell run fills the core (Appendix H): 400 nearly radial orbits with apocentres spread logarithmically between $0.5\, r_c$ and $R_b$, angular momenta 0.05 times that of a circular orbit at the apocentre, random orientations and random orbital phases. They are followed for $300\, r_c/v$, i.e. 12 Gyr. The diagnostic is the time-averaged density of the particles over the last 4 Gyr, and its logarithmic slope at 0.1–$0.5\, r_c$: a population of radial orbits gives a cusp of slope $-2$; a population whose velocities have been fully scattered into all directions, at the same energies, gives a nearly flat core.

*Halting at the front: no.* The exponential sphere has the size $r_c$ and the mass $v^2 r_c/4G$. One passage through it changes the angular momentum by about $\varepsilon(GM_e/r_c)(r_c/v) \sim \varepsilon\, v r_c/4$, whereas a circular orbit at the front needs $vR_b$, i.e. $y = R_b/r_c$ times more, 34–200. In the runs the orbits with apocentres beyond $15\, r_c$ keep $j/j_{\rm circ} = 0.050$ in every case. A distortion that is to halt the shells must have the size of the whole galaxy.

*The core: partly.* Table 2 and Fig. 2 show the results.

**Table 2:** Test orbits in P18 with a triaxial exponential sphere after 12 Gyr. Slope: logarithmic slope of the time-averaged density at 0.1–$0.5\, r_c$. $\beta$: velocity anisotropy inside $r_c$ ($\beta = 1$ for radial orbits, 0 for isotropic ones). $j/j_{\rm circ}$: median for orbits with apocentres below $3\, r_c$ (initially 0.05).

| case | slope | $\beta$ | $j/j_{\rm circ}$ |
|---|---|---|---|
| spherical | $-1.90$ | 0.88 | 0.05 |
| triaxial 0.9/0.8, fixed | $-1.34$ | 0.76 | 0.15 |
| triaxial 0.8/0.6, fixed | $-0.91$ | 0.67 | 0.21 |
| 0.8/0.6, tumbling at $\Omega_p = 0.1\, v/r_c$ | $-1.32$ | 0.71 | 0.18 |
| 0.8/0.6, tumbling at $\Omega_p = 0.3\, v/r_c$ | $-1.49$ | 0.76 | 0.11 |
| 0.9/0.8, tumbling at $\Omega_p = 0.3\, v/r_c$ | $-1.71$ | 0.82 | 0.08 |
| fully scattered, same energies (control) | $-0.39$ | 0.26 | |

- *The mechanism.* A triaxial centre deflects the radial orbits that pass through it and spreads their motion into all directions; the orbits become chaotic (as for triaxial galaxies with a central mass: Gerhard & Binney 1985; Merritt & Valluri 1996). The energies change by only about $0.003\, v^2$: the experiment redistributes angular momentum, it does not heat.

- *How far it gets.* The strongest case covers about two thirds of the way from the radial cusp ($-1.90$) to the fully scattered limit ($-0.39$). Inside about 0.1–$0.2\, r_c$ the profile flattens (Fig. 2), but a core of the size $r_c$ is not formed in 12 Gyr.

- *Tumbling weakens it.* The faster the distortion rotates, the more its pull averages out over an orbit, and the steeper the inner slope stays.

- *Robustness.* Halving the time step and changing the random sample move the slopes by at most 0.07 (`triax_control_out.txt`).

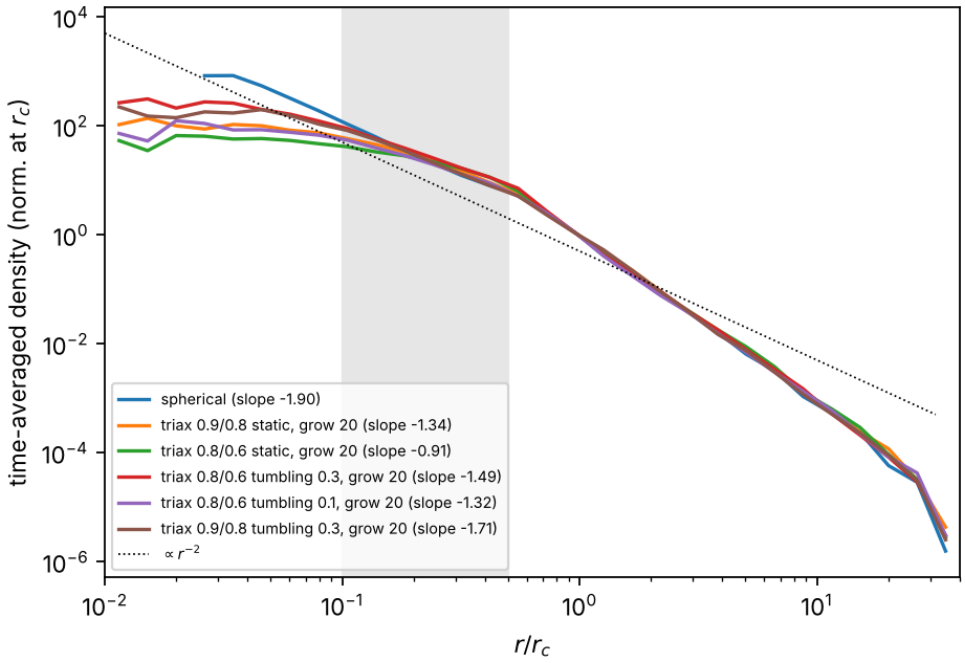



**Figure 2:** Time-averaged density of the nearly radial test orbits in P18 with a triaxial exponential sphere, after 12 Gyr, normalised at $r_c$; the slopes are fitted in the grey band. The spherical case keeps the $r^{-2}$ cusp of radial orbits; the triaxial cases flatten inside about 0.1–0.2 $r_c$. The steep fall beyond $r_c$ only reflects the chosen distribution of apocentres.

*The caveat.* The orbits move in a fixed potential that already contains the core. The experiment therefore tests whether a triaxial core can keep itself cored against radially infalling matter, not how the core forms; and the exponential sphere *is* the core. A full answer needs a self-consistent calculation, an $N$-body run, in which the triaxial shape arises and grows from the infall itself, for example through the radial-orbit instability.

**What the two experiments show.** They separate the two assumptions of the paper. The halt at the front (A3) needs torques on the scale of the whole galaxy and about its own axis, i.e. a triaxial or bar-like distortion of the outer halo during the collapse; a flattening about one axis, or a distortion confined to the centre, cannot do it. The settling into a core (A4), by contrast, can be helped by a triaxial centre: it scatters the radial orbits that would otherwise build a cusp, though in these runs only partly. Both point to the same next step, a three-dimensional collapse of the paper's seed with its own non-spherical dynamics.

# Part II
# Settling into a core

## 4 The question

The paper builds a dwarf galaxy from a seed whose shells share the energy $E = -v^2/2$. Each shell halts at the accretion front, $R_b = 0.175\,vt$, where the galaxy already built encloses its mass (assumption A3). If the halted matter simply stayed where it halted, with $M \propto R_b$, the galaxy would be a *singular isothermal sphere* (SIS),

$$\rho_{\rm SIS} = \frac{v^2}{4\pi G R^2}, \qquad M_{\rm SIS}(R) = \frac{v^2 R}{G}, \qquad v_c = v, \tag{14}$$

with a flat rotation curve all the way to the centre and an infinite central density. The observed dwarfs instead follow P18, which has a core. The paper bridges the gap with assumption (A4): halted matter settles, conserving mass and energy, into P18 with the same $v$. It does not say how. The paper's one-dimensional collisionless calculation (its Appendix H) shows that spherical shells left to themselves do not do it: they form a cusp.

Before asking *how* the settling could happen, one can ask *what* it must achieve. That is the subject of Part II. We compare the SIS and P18 at three levels, each more detailed than the one before:

1. where the mass is (Sect. 5);
2. what orbits the mass can be on: first circular orbits (Sect. 6), then general orbit populations, described by distribution functions (Sect. 7);
3. how tightly the mass is bound (Sect. 8).

Section 9 then spells out what this means for (A4). In Part II we use units $G = v = r_c = 1$, so that masses are in units of $v^2 r_c/G$, energies per unit mass in $v^2$, and angular momenta per unit mass in $v r_c$; the radius is $x = R/r_c$.

# 5 Where the mass is: P18 as a rearranged SIS

## 5.1 The profile

The P18 profile is (paper, Eq. 1)

$$\rho(x) = \frac{v^2 u^2}{4\pi G R^2} = \rho_{\rm SIS}(R)\, u(x)^2, \qquad u = 1 - (1-x)\, e^{-x}, \tag{15}$$

with enclosed mass $M = (v^2 r_c/G)\, m(x)$, where

$$m(x) = \int_0^x u^2 \, \mathrm{d}x' = x(1-e^{-x})^2 + \tfrac{1}{4}\big[1 - (1 + 2x + 2x^2)e^{-2x}\big]. \tag{16}$$

Equation (15) already says the essential thing: *P18 is the SIS multiplied by* $u^2$. All the information about the core is in the factor $u^2$.

## 5.2 A useful rewriting of $u$

Writing $u$ as $1 + (x-1)e^{-x}$ makes its behaviour obvious:

- for $x < 1$ the correction $(x-1)e^{-x}$ is negative: $u < 1$, so P18 has *less* density than the SIS;
- at $x = 1$ it vanishes: $u(1) = 1$ *exactly*, so P18 and the SIS have the same density at $R = r_c$;
- for $x > 1$ it is positive: $u > 1$, P18 has *more* density than the SIS. The correction is largest where its derivative, $(2-x)e^{-x}$, vanishes, at $x = 2$, where $u^2 = (1 + e^{-2})^2 = 1.289$; beyond that it decays exponentially.

The core radius thus has a precise meaning: $r_c$ is the radius at which P18 crosses the singular isothermal sphere of the same $v$ (Fig. 3a).

### 5.3 How much mass is moved

The mass that P18 lacks inside $r_c$, compared with the SIS, is

$$\int_0^1 (1-u^2)\,\mathrm{d}x = 1 - m(1) = 0.520, \tag{17}$$

and the mass it has in addition outside $r_c$ is

$$\int_1^\infty (u^2-1)\,\mathrm{d}x = 0.770. \tag{18}$$

The difference, $0.770 - 0.520 = 0.250$, is the mass excess $c = \int_0^\infty (u^2-1)\,\mathrm{d}x = \frac{1}{4}$ of the paper (Sect. 2.2). The enclosed masses become equal again at $x = 3.29$, where $m(x) = x$ (Table 3).

**Table 3:** Enclosed mass of P18 and of the SIS of the same $v$ (units $v^2 r_c/G$). The last column is the mass that P18 still lacks inside $x$.

| $x = R/r_c$ | $m(x)$ (P18) | $x$ (SIS) | $x - m(x)$ |
|---|---|---|---|
| 0.1 | 0.0012 | 0.1 | 0.099 |
| 0.3 | 0.026 | 0.3 | 0.274 |
| 0.5 | 0.097 | 0.5 | 0.403 |
| 1 | 0.480 | 1 | 0.520 |
| 2 | 1.686 | 2 | 0.314 |
| 3 | 2.943 | 3 | 0.057 |
| 5 | 5.182 | 5 | −0.182 |
| 10 | 10.249 | 10 | −0.249 |

Two remarks help to read these numbers.

- *The rearrangement is small.* A galaxy with core ratio $y = R_b/r_c$ contains the mass $y\,v^2 r_c/G$ inside its front. The mass taken out of the core is therefore a fraction $0.52/y$ of the galaxy: 1.5% for the example galaxy of the paper ($y = 34.4$), 0.5% for $y = 100$ and 0.26% for $y = 200$. The core is a small rearrangement in the centre of an otherwise isothermal galaxy.
- *The mass excess is a compaction.* Far out $m \to x + \frac{1}{4}$, so a P18 galaxy holding the mass $M$ has its front at $R_b = GM/v^2 - r_c/4$: a quarter of a core radius inside the front of the SIS with the same mass. In words: the matter removed from the core is put back just outside it, and a little more, and the galaxy as a whole is slightly more compact.

### 5.4 A remark on the very centre

Near the centre $u = 2x - \frac{3}{2}x^2 + \ldots$, so

$$\frac{\rho}{\rho(0)} = \left(\frac{u}{2x}\right)^2 = 1 - \tfrac{3}{2}x + O(x^2), \qquad \rho(0) = \frac{v^2}{\pi G r_c^2}. \tag{19}$$

The density falls *linearly* with radius: its logarithmic slope goes to zero, so P18 has a core in the usual sense, but its slope $\mathrm{d}\rho/\mathrm{d}R$ does not. The Burkert profile, $\rho \propto 1/[(1+x)(1+x^2)]$, shares this property; the pseudo-isothermal sphere, $\rho \propto 1/(1+x^2)$, does not. We shall meet the consequence in Sect. 7.3.

## 6 Circular orbits: the angular momentum of each shell

The halting rule of the paper places each shell, in the idealisation, on a circular orbit. A circular orbit at $R$ has $v_c^2 = GM(R)/R$ and specific angular momentum

$$j = R v_c = \sqrt{GM(R)\,R} = \frac{GM(R)}{v_c(R)}. \tag{20}$$

In the SIS, $v_c = v$ everywhere and each shell has $j = GM/v$. In P18 the shell enclosing the same mass sits at a different radius, and $j = GM/v_c$ with the smaller circular speed of the core. In units of the SIS value,

$$\frac{j_{\rm P18}}{GM/v} = \frac{v}{v_c(R)} = \sqrt{\frac{x}{m(x)}}, \tag{21}$$

which is 1.44 at $r_c$, 3.4 at $0.3\,r_c$ and 9.2 at $0.1\,r_c$ (Table 4, Fig. 3b). The seed gives each shell, at turnaround, $j = \sqrt{2}\lambda v R_{\rm ta} = 2\sqrt{2}\lambda\, GM/v = 0.099\, GM/v$ for the median spin $\lambda = 0.035$ (paper, Appendix H).

**Table 4:** Angular momentum needed by a shell on a circular orbit in P18.

| $x$ | $v_c/v$ | $j/(GM/v)$ | $j/j_{\rm seed}$ |
|---|---|---|---|
| 0.1 | 0.109 | 9.16 | 92 |
| 0.3 | 0.294 | 3.40 | 34 |
| 0.5 | 0.442 | 2.27 | 23 |
| 1 | 0.693 | 1.44 | 15 |
| 2 | 0.918 | 1.09 | 11 |
| 3 | 0.991 | 1.01 | 10 |
| $\geq 5$ | 1.01–1.02 | 0.98–0.99 | 10 |

Read literally, this says: in the flat part every shell needs ten times the seed's angular momentum, and in the core even more, up to a hundred times. The seed does the opposite: its $j \propto M$ is smallest for the innermost, earliest shells. This is the "ten times too little" of Sect. 2, sharpened in the core.

But this reading assumes circular orbits, and nothing forces them. A rotation curve measures only the enclosed mass, not the orbits that carry it. The same density can be carried by circular orbits, by radial orbits, or by anything in between. To find out which orbits P18 *allows*, we need distribution functions.

# 7 Orbit populations: distribution functions

## 7.1 What a distribution function is

A collisionless system is described by its distribution function $f(\mathbf{x}, \mathbf{v})$: the mass per unit volume of phase space, so that $f\,\mathrm{d}^3x\,\mathrm{d}^3v$ is the mass in a small cell of positions and velocities. The density is $\rho(\mathbf{x}) = \int f\,\mathrm{d}^3v$.

In a steady state, $f$ is constant along every orbit (Liouville's theorem), so it can depend on $\mathbf{x}$ and $\mathbf{v}$ only through quantities that are conserved along orbits. This is the *Jeans theorem*. In a static spherical potential $\Phi(r)$ the conserved quantities are the energy per unit mass $E = \Phi(r) + \frac{1}{2}v^2$ and the angular momentum vector $\mathbf{L}$; for a spherical system without net rotation, $f = f(E, L)$ with $L = |\mathbf{L}|$.

A function of two variables cannot be recovered from a density, which is a function of one. So we must choose a one-parameter family of orbit populations, and ask for which members $f \geq 0$ everywhere. A negative $f$ means a negative mass somewhere in phase space: that population of orbits cannot produce the density. We use two families:

- *isotropic*, $f = f(E)$: velocities point in all directions with equal probability at every point;
- *Osipkov–Merritt* (Osipkov 1979; Merritt 1985), $f = f(Q)$ with $Q = E + L^2/(2r_a^2)$: isotropic well inside the *anisotropy radius* $r_a$, increasingly radial outside it.

The potential is that of P18 itself, $\mathrm{d}\Phi/\mathrm{d}r = GM(r)/r^2 = m(x)/x^2$ in our units, with $\Phi(0) \equiv 0$. Since $m \to x + \frac{1}{4}$, far out $\Phi \to \ln x + 0.134 - 1/(4x)$: the potential grows without bound, every orbit is bound, and $E$ runs from $\Phi(0) = 0$ to $\infty$.

### 7.2 Eddington's inversion, step by step

For an isotropic population at radius $r$, the velocities with speeds between $v$ and $v + \mathrm{d}v$ fill a spherical shell of volume $4\pi v^2 \mathrm{d}v$ in velocity space. With $E = \Phi + \frac{1}{2}v^2$ we have $v\,\mathrm{d}v = \mathrm{d}E$ and $v = \sqrt{2(E-\Phi)}$, so

$$\rho(r) = 4\pi \int_0^\infty f\, v^2 \mathrm{d}v = 4\pi \int_{\Phi(r)}^\infty f(E)\, \sqrt{2(E-\Phi)}\, \mathrm{d}E. \tag{22}$$

The right-hand side depends on $r$ only through $\Phi(r)$. Since $\Phi$ grows monotonically with $r$, we may regard $\rho$ as a function of $\Phi$. Differentiating with respect to $\Phi$ (the boundary term vanishes because $\sqrt{E-\Phi} = 0$ at the lower limit),

$$\frac{\mathrm{d}\rho}{\mathrm{d}\Phi} = -2\sqrt{2}\,\pi \int_\Phi^\infty \frac{f(E)\,\mathrm{d}E}{\sqrt{E-\Phi}}. \tag{23}$$

This is an *Abel integral equation*: the known function $g(\Phi) = -(\mathrm{d}\rho/\mathrm{d}\Phi)/(2\sqrt{2}\pi)$ equals $\int_\Phi^\infty f(E)\,(E-\Phi)^{-1/2}\mathrm{d}E$. Abel showed that it has the unique solution

$$f(E) = -\frac{1}{\pi}\frac{\mathrm{d}}{\mathrm{d}E}\int_E^\infty \frac{g(\Phi)\,\mathrm{d}\Phi}{\sqrt{\Phi-E}}. \tag{24}$$

(You can check it by inserting the equation for $g$ and exchanging the order of integration; the inner integral is $\int_E^{E'}[(\Phi-E)(E'-\Phi)]^{-1/2}\mathrm{d}\Phi = \pi$.) Substituting $g$, and then $\Phi = E + s$ so that the $E$-derivative can be taken inside the integral,

$$\boxed{f(E) = \frac{1}{2\sqrt{2}\,\pi^2}\int_E^\infty \frac{\mathrm{d}^2\rho}{\mathrm{d}\Phi^2}\,\frac{\mathrm{d}\Phi}{\sqrt{\Phi-E}}} \tag{25}$$

This is *Eddington's formula* (Eddington 1916; Binney & Tremaine 2008, Sect. 4.3), in the form suited to a potential without an escape energy. A useful test: for the SIS, $\rho \propto e^{-2\Phi/v^2}$, so $\mathrm{d}^2\rho/\mathrm{d}\Phi^2 = 4\rho/v^4$, and Eq. (25) gives $f \propto e^{-2E/v^2}$: a Maxwellian velocity distribution with dispersion $\sigma^2 = v^2/2$, the same at every radius. This is why the SIS is called isothermal.

For P18 we compute $\mathrm{d}^2\rho/\mathrm{d}\Phi^2$ analytically from Eqs. (15) and (16), using $\mathrm{d}/\mathrm{d}\Phi = (\mathrm{d}\Phi/\mathrm{d}r)^{-1}\mathrm{d}/\mathrm{d}r$, and evaluate Eq. (25) numerically with the substitution $\Phi = E + t^2$, which removes the integrable singularity at $\Phi = E$. As a check, the density rebuilt from $f$ with Eq. (22) agrees with P18 to 0.03% from $10^{-3}$ to $100\,r_c$.

### 7.3 The isotropic P18

The result (Fig. 4a):

- $f(E) > 0$ *everywhere*: P18 can be made of isotropic orbits.
- *Outer energies*: for orbits that live beyond about $4r_c$, $f$ is within 8% of the isothermal $f \propto e^{-2E/v^2}$ of the SIS with the same far field. Outside the core, P18 is an isothermal sphere in phase space as well.
- *Inner energies*: towards the bottom of the potential $f$ rises as

$$f(E) \simeq \frac{0.0105}{E - \Phi(0)} \qquad \big(\text{units } G = v = r_c = 1\big). \tag{26}$$

  This is the phase-space signature of the linear density decline, Eq. (19). Near the centre $\Phi = \frac{2}{3}x^2$, so $\rho \simeq \rho(0)[1 - \frac{3}{2}(\frac{3}{2}\Phi)^{1/2}]$ and $\mathrm{d}^2\rho/\mathrm{d}\Phi^2 = \frac{1}{4}(\frac{3}{2})^{3/2}\rho(0)\,\Phi^{-3/2}$; inserted in Eq. (25), with $\int_E^\infty \Phi^{-3/2}(\Phi-E)^{-1/2}\mathrm{d}\Phi = 2/E$, this gives $f = (\frac{3}{2})^{3/2}\rho(0)/(4\sqrt{2}\pi^2 E)$, i.e. the coefficient 0.0105 found numerically. The divergence is integrable and harmless: the density stays finite.

### 7.4 How radial may the orbits be?

Matter arrives by radial infall, so the natural question is how much of the radial character it may keep. In the Osipkov–Merritt family $f$ depends on $Q = E + L^2/(2r_a^2)$. At radius $r$, with radial and tangential velocities $v_r$ and $v_t$, $L = rv_t$ and

$$Q = \Phi + \tfrac{1}{2}v_r^2 + \tfrac{1}{2}v_t^2\Big(1 + \frac{r^2}{r_a^2}\Big). \tag{27}$$

Stretching the tangential velocities, $v_t' = v_t(1 + r^2/r_a^2)^{1/2}$, turns $Q$ into an ordinary energy, $Q = \Phi + \frac{1}{2}(v_r^2 + v_t'^2)$, while the volume element of velocity space shrinks by the factor $1 + r^2/r_a^2$. Hence

$$\Big(1 + \frac{r^2}{r_a^2}\Big)\rho(r) = 4\pi \int_\Phi^\infty f(Q)\,\sqrt{2(Q-\Phi)}\,\mathrm{d}Q : \tag{28}$$

exactly Eq. (22), with $\rho$ replaced by the "augmented density" $\rho_Q = (1 + r^2/r_a^2)\rho$. So $f(Q)$ follows from Eddington's formula applied to $\rho_Q$. The velocity anisotropy,

$$\beta(r) = 1 - \frac{\sigma_t^2}{2\sigma_r^2} = \frac{r^2}{r^2 + r_a^2}, \tag{29}$$

is 0 (isotropic) at the centre and tends to 1 (purely radial) far outside $r_a$.

Small $r_a$ means radial orbits far in. We find (Fig. 4b):

- for $r_a \geq 0.865\,r_c$ the function $f(Q)$ is positive everywhere;
- for smaller $r_a$ it becomes negative, always at the energies of orbits around the core: for $r_a = 0.7\,r_c$ at energies of circular orbits between 0.44 and $1.3\,r_c$, for $r_a = 0.5\,r_c$ between 0.25 and $1.5\,r_c$.

The most radial P18 of this family thus has $r_a = 0.87\,r_c$, with

$$\beta = 0.57 \text{ at } r_c, \qquad 0.84 \text{ at } 2r_c, \qquad 0.97 \text{ at } 5r_c \tag{30}$$

(Fig. 3c). Outside about $2r_c$ the orbits may be almost purely radial; inside $r_c$ they must be close to isotropic.

This is an instance of a general theorem. For a density $\rho \propto r^{-\gamma}$ at the centre, a positive distribution function requires $\gamma \geq 2\beta(0)$ (An & Evans 2006; proved for wide classes of models by Ciotti & Morganti 2010). A cusp can be built from radial orbits, but a core, $\gamma = 0$, cannot have radially biased orbits at its centre. Purely radial orbits, which all pass through the centre, always produce $\rho \propto r^{-2}$ there; this is why the radial infall of the paper's shell run fills the centre with a cusp.

## 8 How tightly the mass is bound

The distribution function also tells how the mass is distributed in energy. The phase-space volume with energies between $E$ and $E + \mathrm{d}E$ is $g(E)\,\mathrm{d}E$, with the *density of states*

$$g(E) = \int \mathrm{d}^3x \int \mathrm{d}^3v\; \delta\big(E - \Phi - \tfrac{1}{2}v^2\big) = 16\pi^2 \int_0^{r_E} r^2\sqrt{2(E-\Phi(r))}\,\mathrm{d}r, \tag{31}$$

where $\Phi(r_E) = E$; the factor $4\pi v = 4\pi\sqrt{2(E-\Phi)}$ comes from $\mathrm{d}^3v = 4\pi v^2\mathrm{d}v$ and $\mathrm{d}E = v\,\mathrm{d}v$. The mass per unit energy is $N(E) = f(E)\,g(E)$, and the mass bound more tightly than $E$ is $M(<E) = \int^E N\,\mathrm{d}E'$.

To compare P18 with the SIS we must put both on the same energy scale. We take both isotropic and match their potentials far out, where the two galaxies are the same (the SIS potential

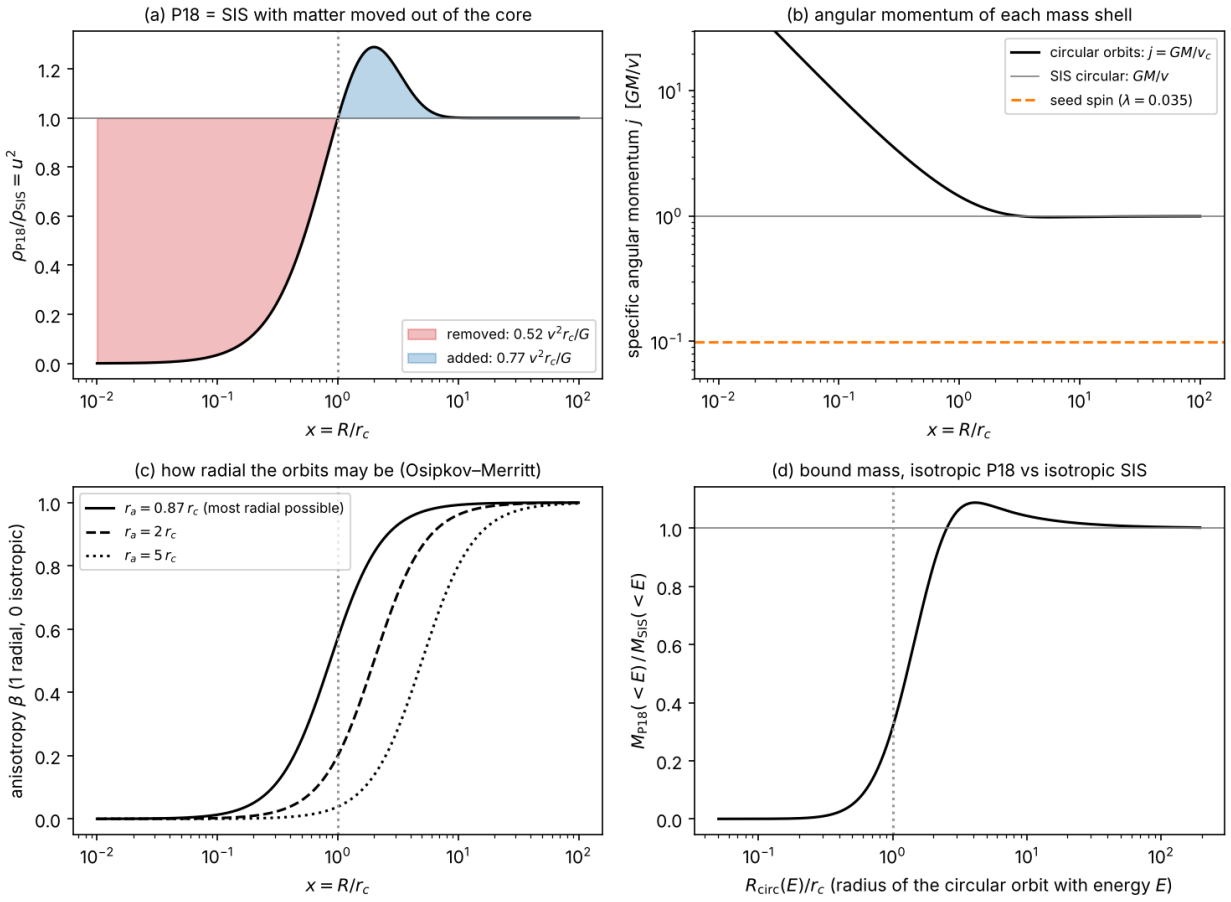



**Figure 3:** What turns the SIS into P18 (units $G = v = r_c = 1$). (a) Density ratio $u^2$: P18 lacks $0.52\,v^2 r_c/G$ inside $r_c$ (red) and has 0.77 more outside (blue). (b) Specific angular momentum of a shell on a circular orbit, in units of $GM/v$, compared with the SIS and with the seed's spin. (c) Velocity anisotropy of Osipkov–Merritt models; $r_a = 0.87\,r_c$ is the most radial one with a positive distribution function. (d) Mass with energy below $E$ in the isotropic P18, relative to the isotropic SIS with the same far field, against the radius of the circular orbit of that energy. Script: `p18_inverse_fig.py`.

is $\ln x + 0.134$). To give each energy a meaning, we label it by the radius $R_{\rm circ}(E)$ of the circular orbit of that energy in P18. Table 5 and Fig. 3d show the result.

The SIS has matter at all energies down to $-\infty$, its infinitely deep centre. P18 has none below $\Phi(0)$, and little just above it: only a quarter as much matter as the SIS is bound more tightly than a circular orbit at $0.9\,r_c$, and 1% as much more tightly than one at $0.3\,r_c$. The missing matter reappears at the energies of orbits at 2.6–30 $r_c$, with up to 9% more than in the SIS. In the language of energy, the core is the *absence of the most bound matter*: the matter that in an SIS would sit deep in the central cusp is, in P18, on orbits of a few core radii.

One caution. The energy distribution depends on the orbits, not only on the density; the comparison here is between two *isotropic* systems. With different orbit populations the numbers change, but not the conclusion that the energies of the inner matter must be raised: no rearrangement of orbits at fixed energies can remove the matter from the bottom of the SIS well, because a particle cannot rise above its own energy.

## 9 What this means for assumption (A4)

Assumption (A4) states that halted matter settles, conserving mass and energy, into P18 with the same $v$. The results above turn this into a list of concrete requirements, and also show what (A4) does *not* require.

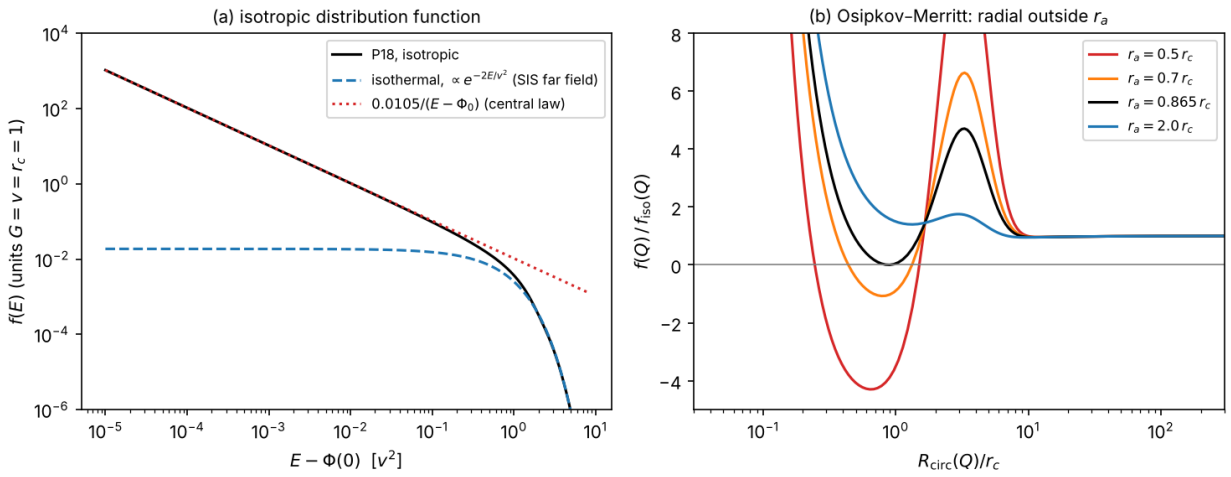



**Figure 4:** (a) Isotropic distribution function of P18 (solid), the isothermal one of the SIS with the same far field (dashed) and the central law $0.0105/(E-\Phi_0)$ (dotted). (b) Osipkov–Merritt distribution functions, relative to the isothermal one, against the radius of the circular orbit of energy $Q$: for $r_a < 0.865\,r_c$ they become negative at the energies of the core. Script: `p18_df_fig.py`.

**Table 5:** Mass bound more tightly than $E$, isotropic P18 and isotropic SIS with the same far field (units $v^2 r_c/G$).

| $E-\Phi(0)$ | $R_{\rm circ}(E)/r_c$ | $M_{\rm P18}(<E)$ | $M_{\rm SIS}(<E)$ | ratio |
|---|---|---|---|---|
| 0.10 | 0.32 | 0.006 | 0.527 | 0.011 |
| 0.30 | 0.62 | 0.060 | 0.641 | 0.094 |
| 0.51 | 0.90 | 0.200 | 0.792 | 0.25 |
| 1.00 | 1.61 | 0.904 | 1.294 | 0.70 |
| 1.50 | 2.58 | 2.15 | 2.13 | 1.01 |
| 1.98 | 4.0 | 3.76 | 3.45 | 1.09 |
| 2.98 | 10.6 | 9.77 | 9.39 | 1.04 |
| 3.98 | 28.6 | 25.9 | 25.6 | 1.015 |
| 5.99 | 211 | 189.7 | 189.3 | 1.002 |

## 9.1 What (A4) requires

1. *A small rearrangement of mass.* Compared with the isothermal galaxy the halted matter would form, $0.52\,v^2 r_c/G$ must be moved from inside $r_c$ to between $r_c$ and about $5r_c$, together with a compaction of the whole galaxy by $r_c/4$ (the mass excess). This is 0.3–1.5% of the galaxy's mass for core ratios $y = 34$–$200$. (A4) concerns the innermost percent of the galaxy, not the galaxy as a whole.

2. *Isotropic orbits in the core.* Inside $r_c$ the orbits must be close to isotropic: in the Osipkov–Merritt family $\beta \le 0.57$ at $r_c$ and $\beta \to 0$ at the centre, and by the theorem of An & Evans no positive distribution function has radially biased orbits at the centre of a core. Radial infall delivers the opposite, $\beta \simeq 1$. The infalling matter must therefore have its velocities turned into all directions *before or while* it passes through the core. The angular momentum this needs is of the core's own scale, $j \sim v r_c$, not $GM/v$ at the front: the inner matter must miss the centre by about a core radius.

3. *An energy lift.* The most bound matter of the isothermal galaxy must be raised by energies of order $v^2$ (Table 5: from below the circular energy at $0.3\,r_c$ to that of orbits at a few $r_c$, a difference of more than $v^2$). A collisionless particle changes its energy only through a potential that changes in time: along its orbit $\mathrm{d}E/\mathrm{d}t = \partial\Phi/\partial t$. A static distortion, however strong, can turn velocities and redistribute angular momentum but cannot lift energies. (A4) therefore

requires a time-dependent potential in the centre: a collapse that is still violently relaxing, a tumbling or oscillating bar, fluctuations driven by gas, or sinking clumps.

4. *Continuity during growth.* During growth the core grows as $r_c \propto t$, and the mass that must be held out of the core, $0.52\,v^2 r_c/G$, grows with it. The rearrangement is not a single event: matter accreted earlier, which settled when the core was smaller, must be lifted again and again as the core sweeps over it. This is the flow of matter in a growing core described in the paper's Appendix I, now with its energy requirement attached.

5. *Mostly an internal transfer of energy.* (A4) adds no energy from outside: the settled galaxy has the energy of the seed. The paper's core equation does this global bookkeeping; for the example galaxy the core term is $-0.0082$ of the binding energy $\frac{1}{2}v^2 M$ and balances the spin ($+0.0037$) and $\Lambda$ ($+0.0045$) terms (paper, Table 2). The local lift of item 3 is larger: of order $0.5\,v^2 r_c/G$ of matter raised by about $v^2$, i.e. some $0.5\,v^4 r_c/G$, about 3% of the binding energy $\frac{1}{2}v^2 \cdot y\,v^2 r_c/G$ for $y = 34$. The net energy of the core is thus only a part of what the inner matter must gain; the rest must come from matter elsewhere in the galaxy that becomes more bound. This is the "internal transfer, not net" of the paper's Table 2, here traced to where it must go. The estimate is rough; a mechanism should be judged by whether it moves energy *inwards to the core matter* at about this rate.

### 9.2 What (A4) does not require

- *Circular orbits at the front.* The flat part of P18, outside about $2r_c$, can be carried by nearly radial orbits (Sect. 7.4). The "tenfold" angular momentum of Table 4 is the price of the idealised picture with circular orbits, not of the density. For the density, what matters outside the core is the distribution of the orbits' energies, i.e. where they turn around, not their angular momenta. Whether the seed delivers exactly the energies for $\rho \propto r^{-2}$ is a separate question; the paper's shell run says: not quite, its curve falls from $v_c^2 = 1.54\,v^2$ at $0.1\,R_b$ to $0.86\,v^2$ at $R_b$ (Appendix H).

- *A net rotation.* Isotropy needs angular momenta in all directions, which cancel; the net spin can stay at the seed's $\lambda \simeq 0.035$.

- *Any change of the outer galaxy.* Beyond about $7r_c$ P18 and the SIS agree to about 1% in density ($u^2 - 1 \simeq 2(x-1)e^{-x}$), and in phase space the isotropic P18 is within 8% of the isothermal distribution function for orbits beyond about $4r_c$.

### 9.3 The candidate mechanisms, measured against these requirements

- *Violent relaxation of a non-spherical collapse.* It provides both the time-dependent potential (item 3) and the turning of velocities (item 2). In exact spherical symmetry it fails, as the paper's shell run shows: the angular momenta are conserved and the radial orbits fill the centre. With the radial-orbit instability, which makes a cold collapse bar-like or triaxial, it is the most natural candidate.

- *A triaxial centre.* The test-particle experiment of Sect. 3 gave isotropy only in part ($\beta = 0.67$ inside $r_c$, against $\beta \leq 0.57$ at $r_c$ and $\beta \to 0$ at the centre required here) and almost no energy change ($|\Delta E| \simeq 0.003\,v^2$), because its potential was static apart from a slow growth. Its partial flattening of the cusp, to a slope of $-0.9$, is what items 2 and 3 lead one to expect from turning velocities without lifting energies. A tumbling or growing distortion that changes on the dynamical time of the core, about 200 Myr in the example galaxy, would act on both.

- *Sinking clumps.* Substructure that sinks by dynamical friction transfers its orbital energy to the central dark matter (El-Zant, Shlosman & Hoffman 2001): an energy lift (item 3) paid from within the galaxy (item 5).

- *Feedback.* Repeated gas outflows make the central potential fluctuate and lift the central dark matter (Pontzen & Governato 2012): items 2 and 3. Its difficulty in the model is the scaling with $v$ (paper, Sect. 5.4; Sect. 2).

- *Self-interacting dark matter.* Collisions isotropise the velocities and conduct energy outwards into the core: items 2 and 3 at once, and the result is close to isothermal, as P18 is outside the core. Its core scales with $v$ in a way set by the cross-section (paper, Sect. 8.2).

### 9.4 What a simulation should look for

These requirements can be checked directly in a simulation of the paper's seed, or in any simulation of a cored dwarf:

- the velocity anisotropy of the dark matter: close to isotropic inside $r_c$, allowed to be radial beyond about $2r_c$;

- the energy distribution $M(< E)$ of the inner matter, compared with Table 5;

- the time dependence of the central potential while the core grows, and whether the energy gained by the inner matter matches the energy lost by the matter just outside.

## 10 Limits of this note

- Part I follows the seed only until each shell halts; the halting itself is the assumption (A3), and a one-dimensional collisionless calculation in which the shells are not halted forms a cusp instead (paper, Sect. 8.1 and Appendix H). The test-particle experiments of Sect. 3 use fixed potentials and are not self-consistent.

- Part II concerns only the *end state*: what P18 is in phase space and how it differs from the SIS. It shows what a settling mechanism must achieve, not that the seed's collapse achieves it.

- The potential is that of P18 alone. In real dwarfs the baryons contribute a median 24% of $V^2$ at $r_c$ (paper, Appendix E); for the total-matter profile of the paper this does not matter, but the orbits of the dark matter alone live in the total potential.

- Two families of distribution functions were used. Others exist; the theorem $\gamma \geq 2\beta(0)$ and the energy argument of Sect. 8 hold beyond them, the number $r_a \geq 0.87\, r_c$ does not.

- The energy comparison of Sect. 8 assumes isotropy for both systems; the statement that inner energies must be raised does not.

## References


An J. H., Evans N. W., 2006, ApJ 642, 752
Bertschinger E., 1985, ApJS 58, 39
Binney J., Tremaine S., 2008, Galactic Dynamics, 2nd ed., Princeton Univ. Press
Ciotti L., Morganti L., 2010, MNRAS 408, 1070
Eddington A. S., 1916, MNRAS 76, 572
El-Zant A., Shlosman I., Hoffman Y., 2001, ApJ 560, 636
Gerhard O. E., Binney J., 1985, MNRAS 216, 467
Governato F., Brook C., Mayer L., et al., 2010, Nature 463, 203
Lynden-Bell D., 1967, MNRAS 136, 101
MacMillan J. D., Widrow L. M., Henriksen R. N., 2006, ApJ 653, 43
Merritt D., 1985, AJ 90, 1027

Merritt D., Valluri M., 1996, ApJ 471, 82
Navarro J. F., Eke V. R., Frenk C. S., 1996, MNRAS 283, L72
Osipkov L. P., 1979, Pis'ma Astron. Zh. 5, 77
Pontzen A., Governato F., 2012, MNRAS 421, 3464
Read J. I., Gilmore G., 2005, MNRAS 356, 107

## Files in this package

| | |
|---|---|
| *Part I* | |
| `halt.py`, `halt_numbers.py` | Sect. 2: orbits with the seed's and with circular angular momentum |
| `sf_numbers.py` | Sect. 2: densities at the front, energies of feedback |
| `axi_test.py`, `axi_energy.py` | Sect. 3: flattening about one axis |
| `triax_core.py`, `triax_control.py` | Sect. 3: triaxial exponential sphere and controls |
| `triax_density_check.py`, `triax_fig.py` | Sect. 3: density of the triaxial potential; its figure |
| *Part II* | |
| `p18core.py` | P18 functions $u$, $m$, $\rho$, accurate at all radii |
| `p18_inverse.py` | Sects. 5–6: mass rearrangement, circular-orbit angular momenta |
| `p18_df.py` | Sects. 7–8: isotropic $f(E)$, density check, $M(< E)$ |
| `p18_om.py` | Sect. 7: Osipkov–Merritt models, smallest $r_a$ |
| `p18_iso_compare.py` | Sect. 7: $f$ against the isothermal one |
| `p18_inverse_fig.py`, `p18_df_fig.py` | the two figures of Part II |
| `*_out.txt` | outputs of the scripts |